AuxiCompass

Converter fails mid-section. The standby does not take over. What now?

The loco pilot gets no guide: what to switch off, how many minutes are left, whether to stop.



Sheds told RDSO the load did not shift from SIV1 to SIV2 (Siemens 2x130 kVA) or after an MCB trip (Medha 2x130 kVA). RDSO minutes, 29 May 2025.

Failures are frequent, and the evidence disappears

~27

failures per 100 converters a year

582 failures in 2,116 units, FY 2024-25, conventional electric locos. The total is our arithmetic from RDSO's table.

The standby does not always take over

- Siemens 2x130 kVA: load did not shift from SIV1 to SIV2 after a fault.
- Medha 2x130 kVA: load did not shift when the front-panel MCB tripped.

The fault log is gone before the shed reads it

20 days

of fault log kept

60 days

between scheduled shed visits

Source: RDSO minutes on 180 kVA SIV failures, meeting of 29 May 2025, Annexure-I. Conventional electric locos, not 3-phase locos.

Monitoring exists. A crew advisor does not

What exists today

- **REMMLOT remote monitoring**
  reports loco health to the shed
- **Loco thermal protection**
  derates or trips the motor
- **CLW analytics spec, 3-phase locos**
  April 2024, deployment unknown

What the crew still lacks

What do I switch off?

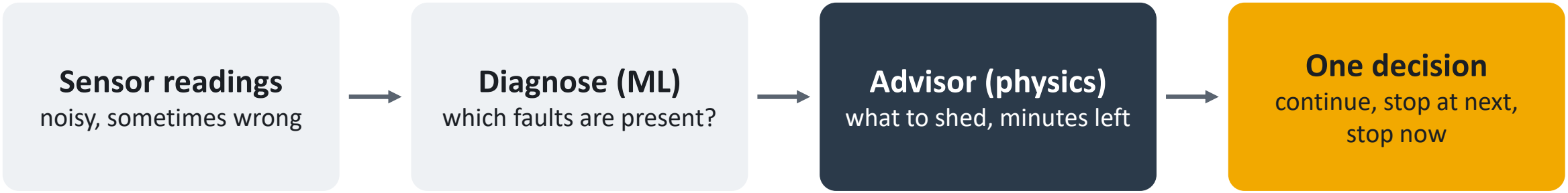
How many minutes do I have?

Continue, or stop?

We searched and found no crew-side advisor. We say "we found none", not "none exists". Sources: IRIMEE slides, CLW documents (medium trust).

Our Solution
Two parts: an advisor, and the proof that it works

1 THE ADVISOR



2 THE PROOF

Fault-injection harness

8 faults > 36 single and double combinations x 4 conditions x 2 policies = 288 runs

Four safety properties checked in every run. The results become the coverage map.

Physics first. ML names the fault; a learned estimate may lower the minutes left, never raise them. Sources: CONTRACT.md sections 1 and 4, docs/model.md.

Model result: 44 unsafe without the advisor, 0 with it

44 / 144

runs unsafe without the advisor

0 / 144

runs unsafe with the advisor

8

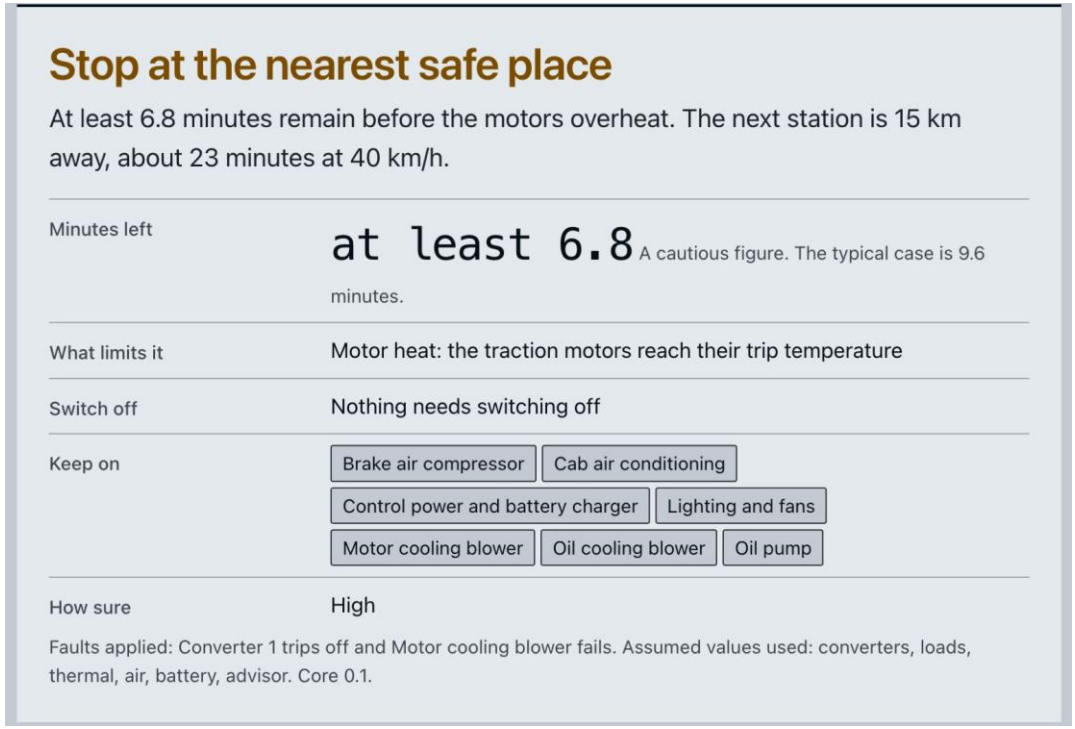
cautious stops the baseline did not need

0

safety properties failed

What the pilot sees: the live app

LIVE DEMO



Model with assumed parameters; harness advisor told the true faults (coverage.json, core 0.1, seed 1). Auto-diagnose, held out and simulated: exact fault set 55%, stuck sensor 100% (models/metrics.json).

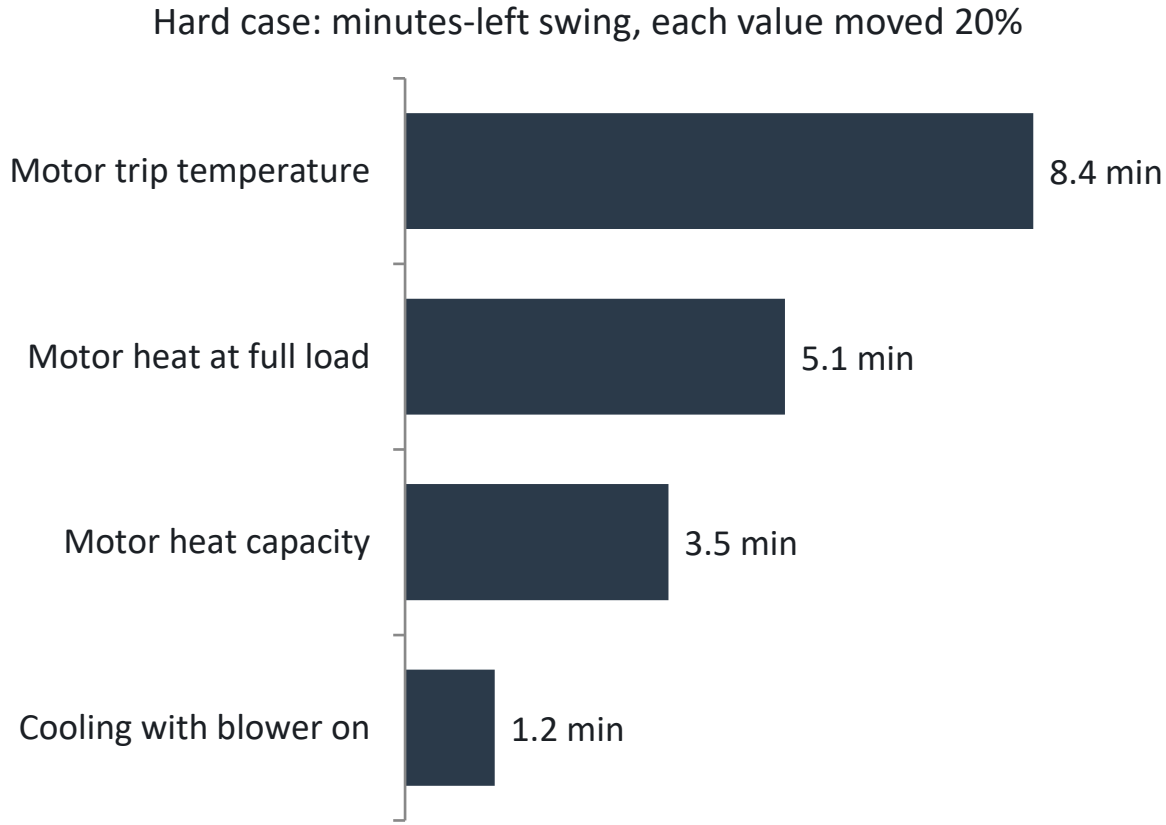
What we left out, and what we assumed

Future Scope

- Failure prediction (no data to learn it from)
- Passenger features
- Any safety-certification claim

ASSUMED

Every loco value in our model is a team assumption, and the app shows this tag. The 0 unsafe result is from our model, with the advisor told the true faults.



Most sensitive assumptions: the motor trip temperature (assumed 180 C) and the brake-air minimum (48.0 min swing for a failed compressor). Leak rate from PLW dispatch test sheets. Source: demo_outputs/sensitivity.json.

From prototype to the loco cab

1. Replace simulated faults with the converter makers' real fault codes
2. Offer the harness to RDSO as an acceptance test for changeover software
3. Model BLW's 3-phase locos: 3 converters sharing load at about 37 Hz
4. Keep 60 days of fault log, not 20, so the shed sees the evidence

Who adopts it: RDSO for the test; zonal sheds and **crews** for the advisor.

What it needs: maker fault codes, shed data, confirmed loco values, an approval route.

AI disclosure: We used Claude, an AI assistant, to research sources and help write code. The running tool contains no language model. We wrote our own tests and marked every assumed value.